

Below is our HOME IP Address and Default Gateway:

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :

IPv4 Address. : 10.0.0.177

Subnet Mask : 255.255.255.0

Default Gateway : 10.0.0.1

What IP Address and Gate Way IP Address should we use inside Windows Servers (DC, Node1 and Node2) to be added to domain windows server hosted on a VMWare workstation without IP Conflict?

C:\Users\DELL>ipconfig

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 1:

Media State : Media disconnected

Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 10:

Media State : Media disconnected

Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :

IPv4 Address. : 10.0.0.177

Subnet Mask : 255.255.255.0

Default Gateway : 10.0.0.1

C:\Users\DELL>

Assume that your **home network is in the 10.0.0.0/24 range**, specifically:

- **Home IP:** 10.0.0.177
- **Home Gateway:** 10.0.0.1
- **Subnet:** 255.255.255.0

 Goal Expectation

You want to:

- Run **Windows Server VMs** (DC, Node1, Node2) in **VMware Workstation**
- Join all to a domain
- Avoid **IP conflicts** with your home network

 Recommended Network Type in VMware Workstation

To avoid interference or IP conflicts with your home network:

Option	Description	Internet Access	Isolation	Best For
Host-Only	VMs can talk to each other & host only	✗	✓ Full	Lab/Domain
NAT	VMs share host's internet via NAT	✓	Partial	Internet in lab
Bridged	VMs join home network directly	✓	✗	Not recommended here

✅ **Recommendation:** Use **Host-Only** network (e.g., VMnet2) for **fully isolated domain lab**.

✅ **Custom Subnet Plan (Avoiding Conflict)**

Since your home network uses 10.0.0.0/24, pick a **completely different range**, e.g.:

Use: 192.168.100.0/24

Subnet Mask: 255.255.255.0

Machine	IP Address	Subnet Mask	Gateway	DNS	Notes
DC	192.168.100.10	255.255.255.0	192.168.100.1 (<i>dummy</i>)	192.168.100.10	Runs AD DS & DNS
Node1	192.168.100.11	255.255.255.0	192.168.100.1	192.168.100.10	Domain-joined
Node2	192.168.100.12	255.255.255.0	192.168.100.1	192.168.100.10	Domain-joined

💡 Gateway is optional in Host-Only. Use a dummy IP (e.g., 192.168.100.1) for consistency if no internet is needed.

⚙️ How to Set This Up

1. VMware Workstation – Create Host-Only Network

- Go to: **Edit > Virtual Network Editor**
- Create or modify a **Host-Only** network (e.g., VMnet2)
 - Subnet IP:** 192.168.100.0
 - Subnet Mask:** 255.255.255.0
 - Disable **DHCP**
 - Leave **host virtual adapter** enabled if you want your host to manage VMs

2. Assign VMs to This Network

- In VM settings (each VM):
 - Go to **Network Adapter**
 - Select **Custom: VMnet2** (or your chosen host-only network)

3. Set Static IPs Inside Each Windows Server VM

Inside each VM:

On DC:

IP Address: 192.168.100.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.100.1 (dummy)

DNS: 127.0.0.1 or 192.168.100.10

On Node1:

IP Address: 192.168.100.11

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.100.1

DNS: 192.168.100.10

On Node2:

IP Address: 192.168.100.12

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.100.1

DNS: 192.168.100.10

✅ All VMs will resolve the domain via DNS on the DC and communicate privately.

🔧 Optional: Internet Access

If you **need internet access** in the domain lab:

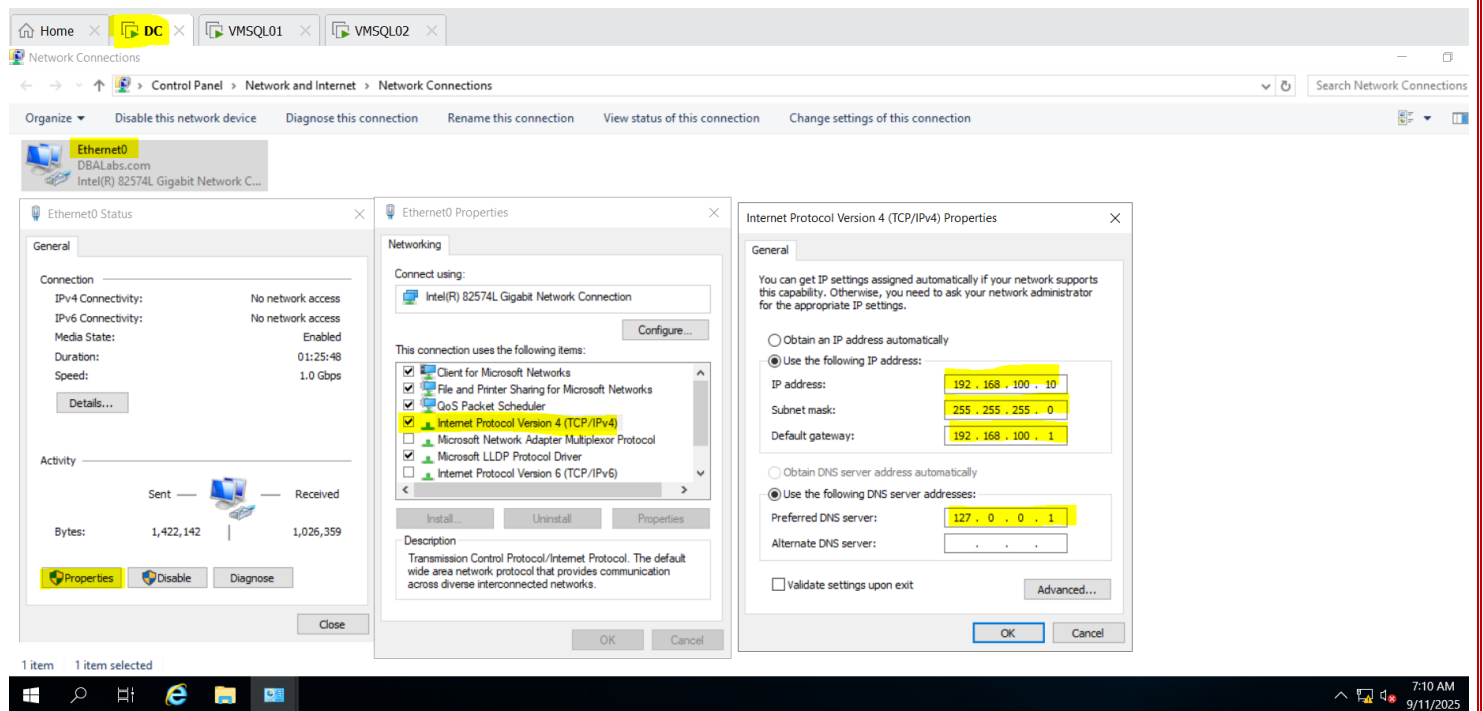
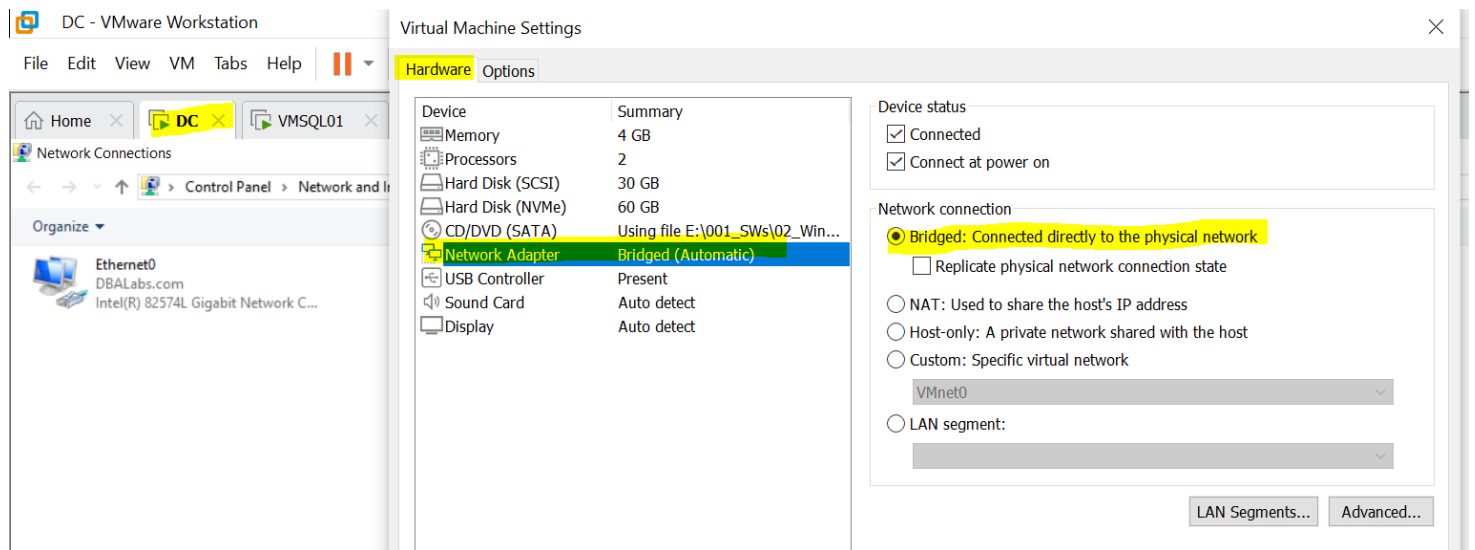
- Use **NAT network** instead (e.g., 192.168.200.0/24)
- Or **dual NICs**:
 - One adapter on NAT for internet
 - One on Host-Only for internal domain communication

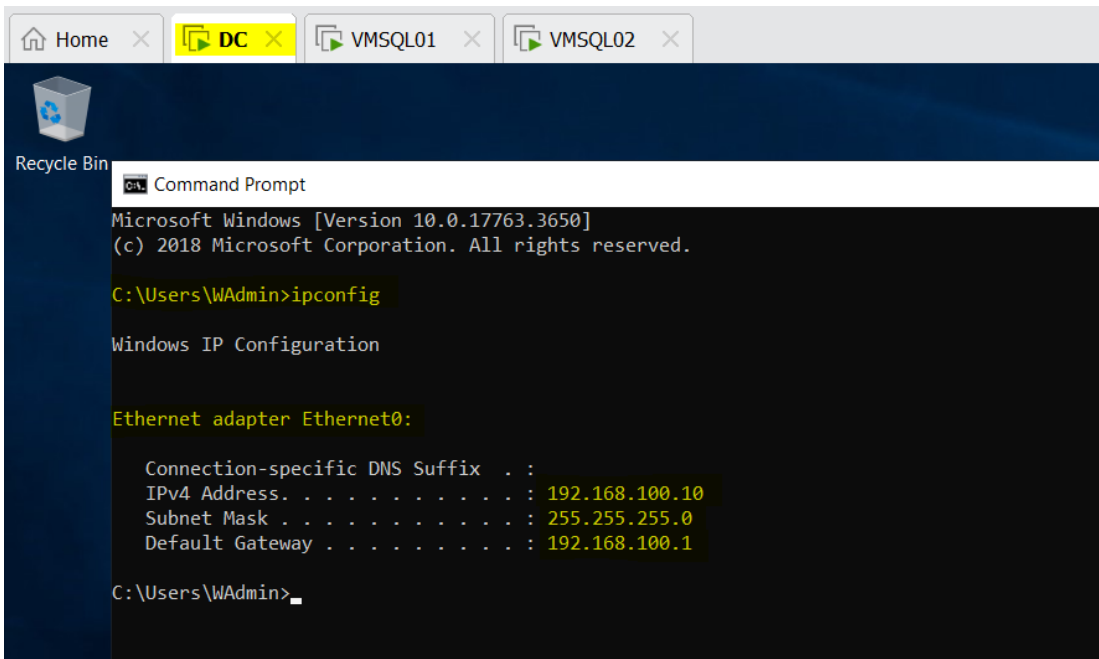
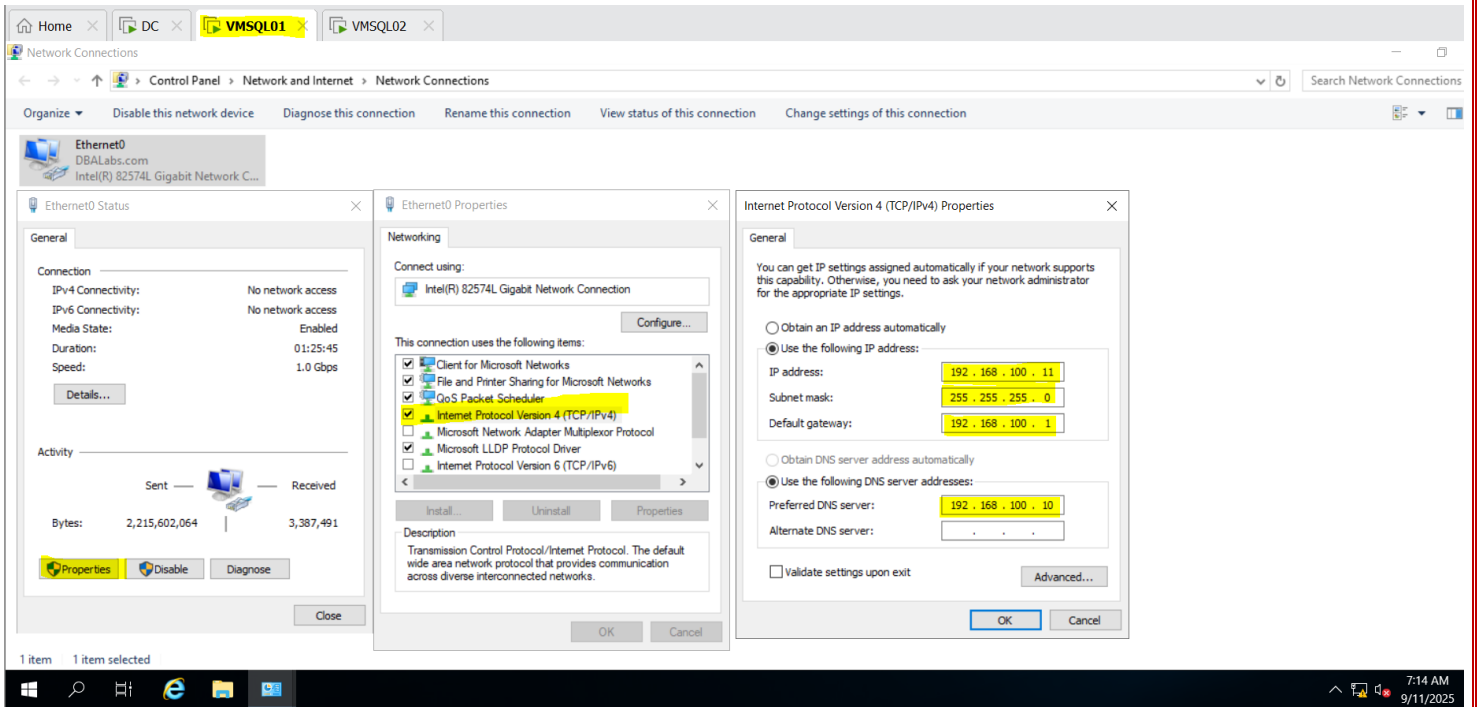
But for most domain testing, **isolated (host-only)** is ideal.

✅ Final Notes

- **Do NOT** use 10.0.0.x in your VMs — that's your home network, will cause conflicts.
- Keep **DNS set to your Domain Controller** — Windows domains rely on DC for name resolution.

At DC: (Domain Controller)





```
Home x DC x VMSQL01 x VMSQL02 x
C:\Users\WAdmin>ping vmsql01

Pinging vmsql01.DBALabs.com [192.168.100.11] with 32 bytes of data:
Reply from 192.168.100.11: bytes=32 time<1ms TTL=128
Reply from 192.168.100.11: bytes=32 time<1ms TTL=128
Reply from 192.168.100.11: bytes=32 time<1ms TTL=128
Reply from 192.168.100.11: bytes=32 time<1ms TTL=128

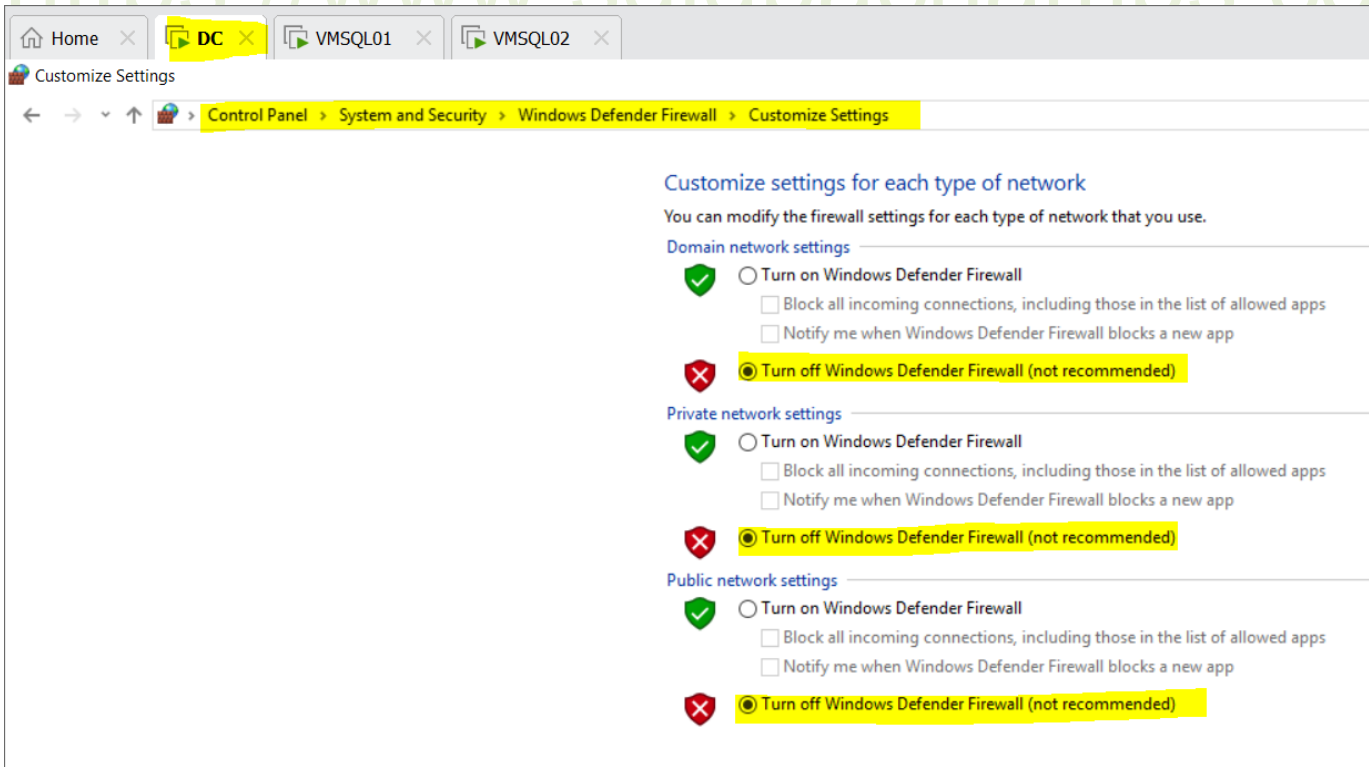
Ping statistics for 192.168.100.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\WAdmin>ping vmsql02

Pinging vmsql02.DBALabs.com [192.168.100.12] with 32 bytes of data:
Reply from 192.168.100.12: bytes=32 time<1ms TTL=128
Reply from 192.168.100.12: bytes=32 time=2ms TTL=128
Reply from 192.168.100.12: bytes=32 time=1ms TTL=128
Reply from 192.168.100.12: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.100.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 1ms

C:\Users\WAdmin>
```



The screenshot shows the Windows Control Panel with the 'System' window open. The 'System' window displays basic information about the computer, including the Windows edition (Windows Server 2019 Datacenter Evaluation), system type (64-bit Operating System, x64), and computer name (DC). The 'System Properties' dialog box is also open, showing the 'Remote' tab. In the 'Remote Desktop' section, the option 'Allow remote connections to this computer' is selected. The 'Remote Assistance' section has the option 'Allow Remote Assistance connections to this computer' unchecked. The 'Advanced' button is visible in the 'Remote Assistance' section.

Control Panel Home

- Device Manager
- Remote settings
- Advanced system settings

View basic information about your computer

Windows edition

Windows Server 2019 Datacenter Evaluation
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System

Processor: 11th Gen Intel(R) Core(TM)
Installed memory (RAM): 4.00 GB
System type: 64-bit Operating System, x64
Pen and Touch: No Pen or Touch Input is available

Computer name, domain, and workgroup settings

Computer name: DC
Full computer name: DC.DBALabs.com
Computer description:
Domain: DBALabs.com

System Properties

Computer Name Hardware Advanced Remote

Remote Assistance

☐ Allow Remote Assistance connections to this computer

Advanced...

Remote Desktop

Choose an option, and then specify who can connect.

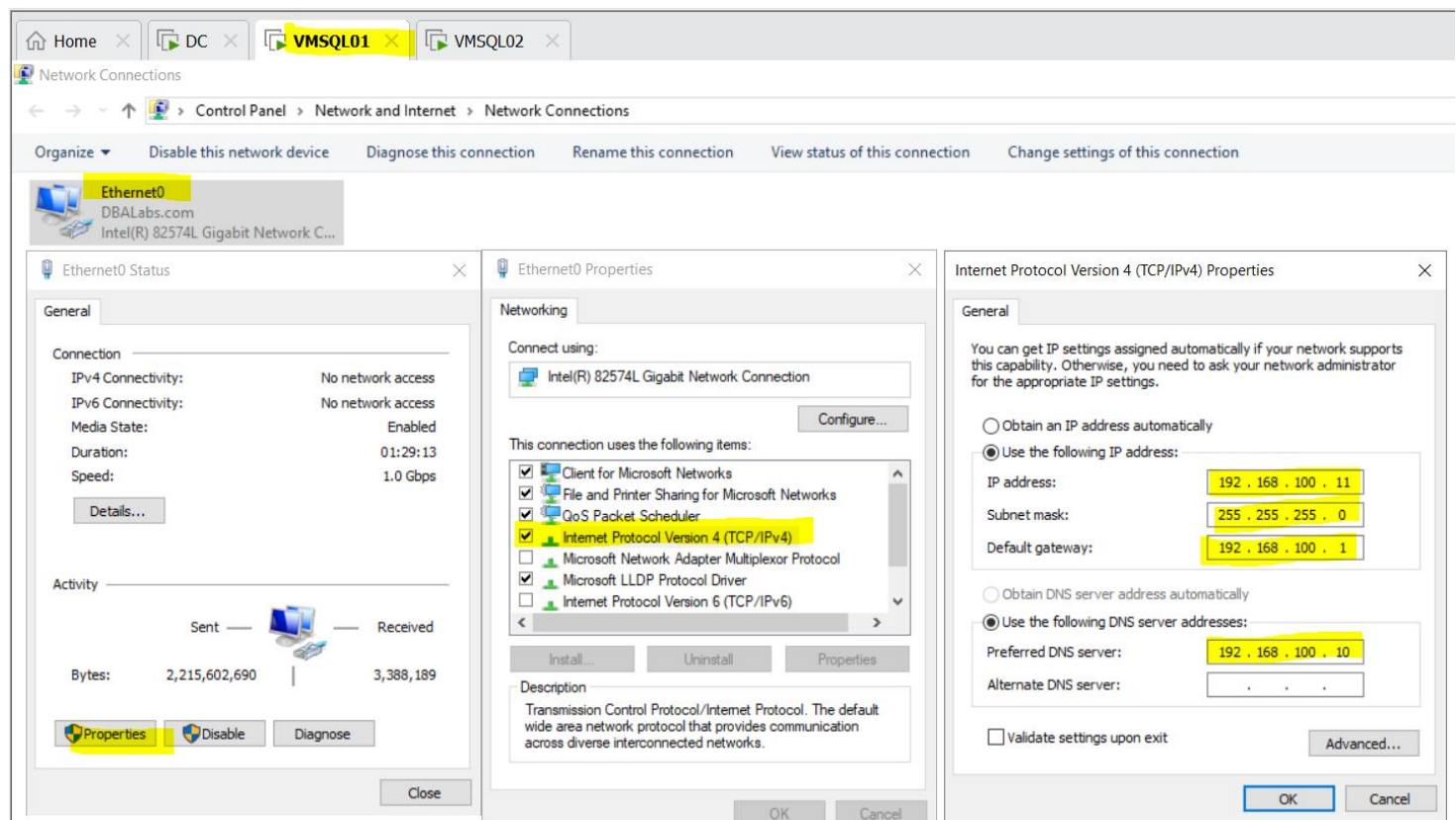
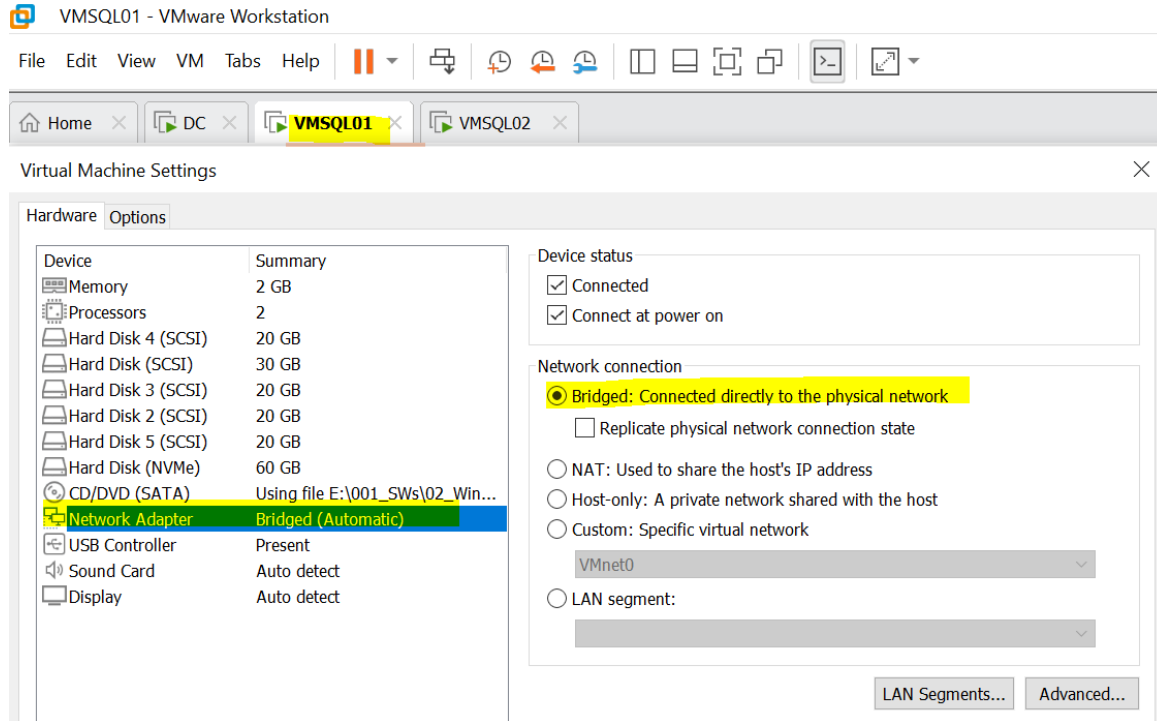
☐ Don't allow remote connections to this computer

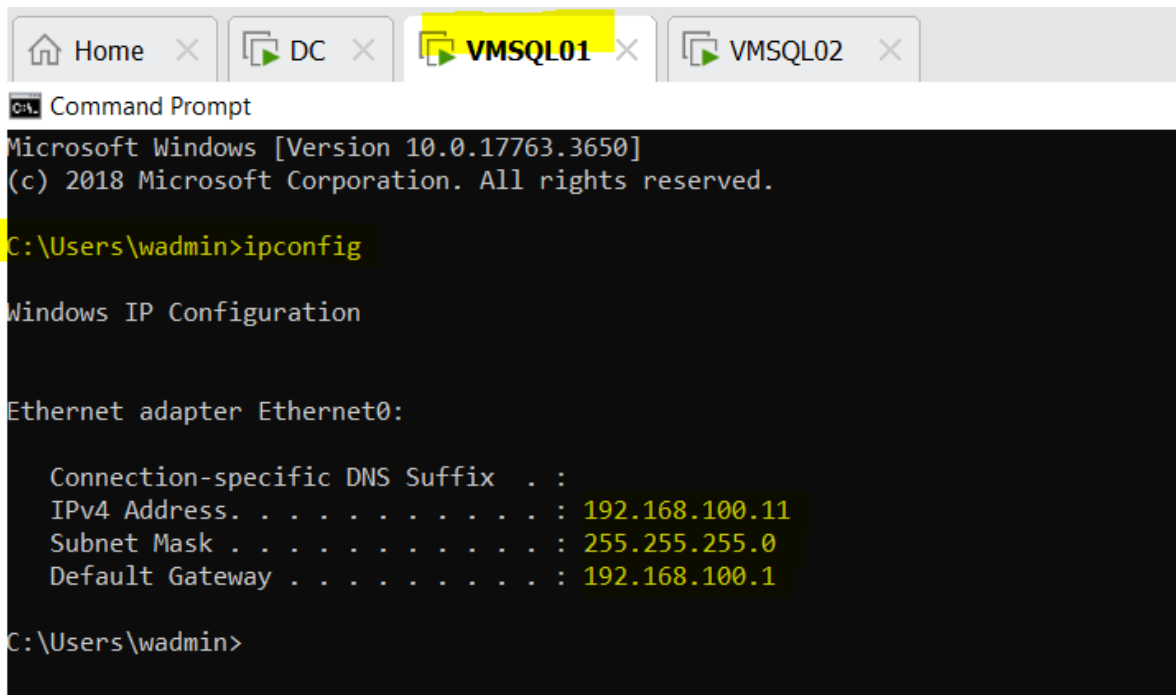
☒ Allow remote connections to this computer

☐ Allow connections only from computers running Remote Desktop with Network Level Authentication (recommended)

Help me choose

Select Users...

At Node1 (VMSQL01):



```
Home x DC x VMSQL01 x VMSQL02 x
C:\ Command Prompt
Microsoft Windows [Version 10.0.17763.3650]
(c) 2018 Microsoft Corporation. All rights reserved.

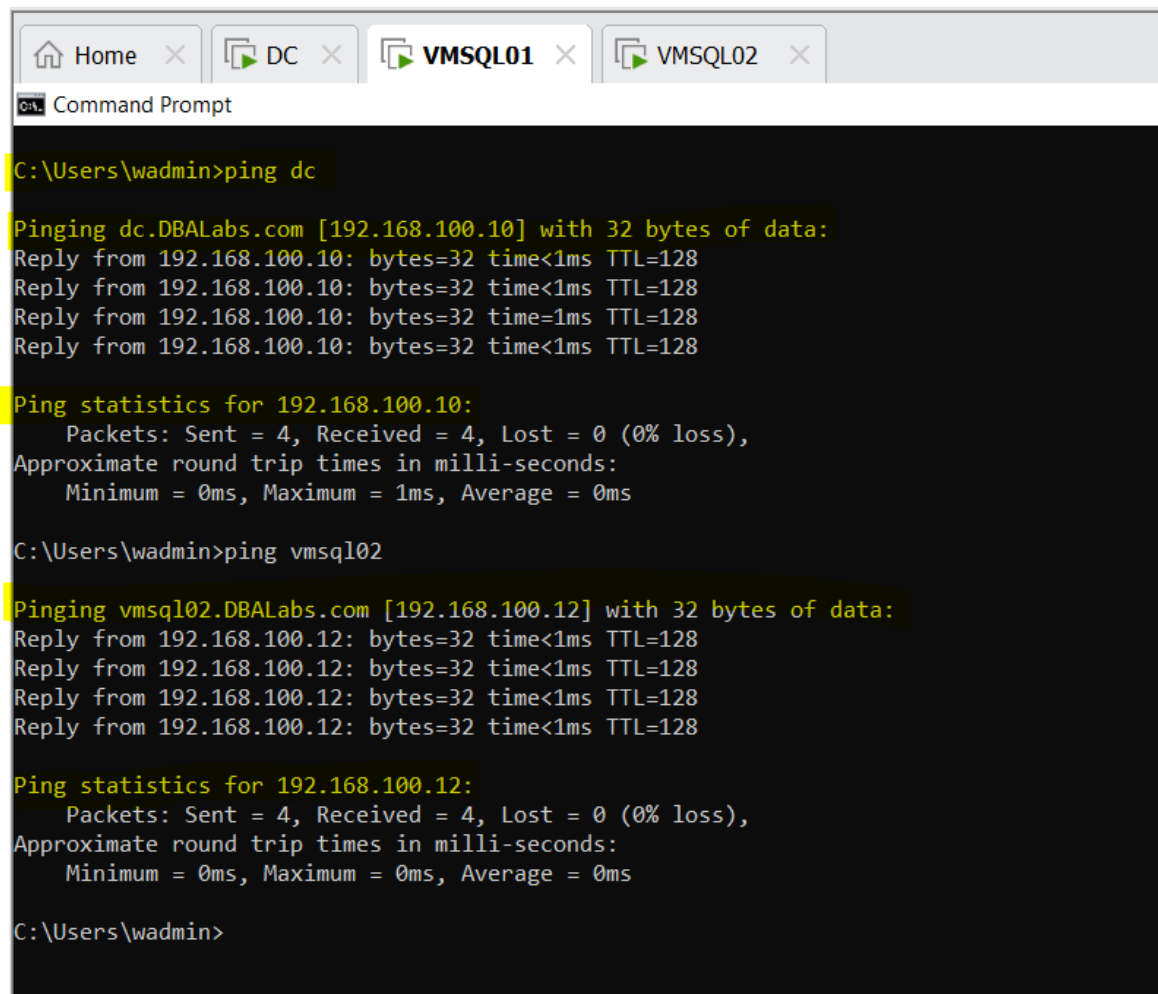
C:\Users\wadmin>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : 
    IPv4 Address. . . . . : 192.168.100.11
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.100.1

C:\Users\wadmin>
```



```
Home x DC x VMSQL01 x VMSQL02 x
C:\ Command Prompt

C:\Users\wadmin>ping dc

Pinging dc.DBALabs.com [192.168.100.10] with 32 bytes of data:
Reply from 192.168.100.10: bytes=32 time<1ms TTL=128
Reply from 192.168.100.10: bytes=32 time<1ms TTL=128
Reply from 192.168.100.10: bytes=32 time=1ms TTL=128
Reply from 192.168.100.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.100.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\wadmin>ping vmsql02

Pinging vmsql02.DBALabs.com [192.168.100.12] with 32 bytes of data:
Reply from 192.168.100.12: bytes=32 time<1ms TTL=128
Reply from 192.168.100.12: bytes=32 time<1ms TTL=128
Reply from 192.168.100.12: bytes=32 time<1ms TTL=128
Reply from 192.168.100.12: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.100.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\wadmin>
```

Home × DC × VMSQL01 × VMSQL02 ×

Customize Settings

Control Panel > System and Security > Windows Defender Firewall > Customize Settings

Customize settings for each type of network

You can modify the firewall settings for each type of network that you use.

Domain network settings

☒ Turn on Windows Defender Firewall

☐ Block all incoming connections, including those in the list of allowed apps

☐ Notify me when Windows Defender Firewall blocks a new app

☒ Turn off Windows Defender Firewall (not recommended)

Private network settings

☒ Turn on Windows Defender Firewall

☐ Block all incoming connections, including those in the list of allowed apps

☐ Notify me when Windows Defender Firewall blocks a new app

☒ Turn off Windows Defender Firewall (not recommended)

Public network settings

☒ Turn on Windows Defender Firewall

☐ Block all incoming connections, including those in the list of allowed apps

☐ Notify me when Windows Defender Firewall blocks a new app

☒ Turn off Windows Defender Firewall (not recommended)

Home × DC × VMSQL01 × VMSQL02 ×

System

Control Panel > System and Security > System

View basic information about your computer

Windows edition

Windows Server 2019 Datacenter Evaluation

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System

Processor: 11th Gen Intel(R) Core(TM)

Installed memory (RAM): 2.00 GB

System type: 64-bit Operating System, x64

Pen and Touch: No Pen or Touch Input is available

Computer name, domain, and workgroup settings

Computer name: VMSQL01

Full computer name: VMSQL01.DBALabs.com

Computer description:

Domain: DBALabs.com

System Properties

Computer Name | Hardware | Advanced | Remote

Remote Assistance

☐ Allow Remote Assistance connections to this computer

Advanced...

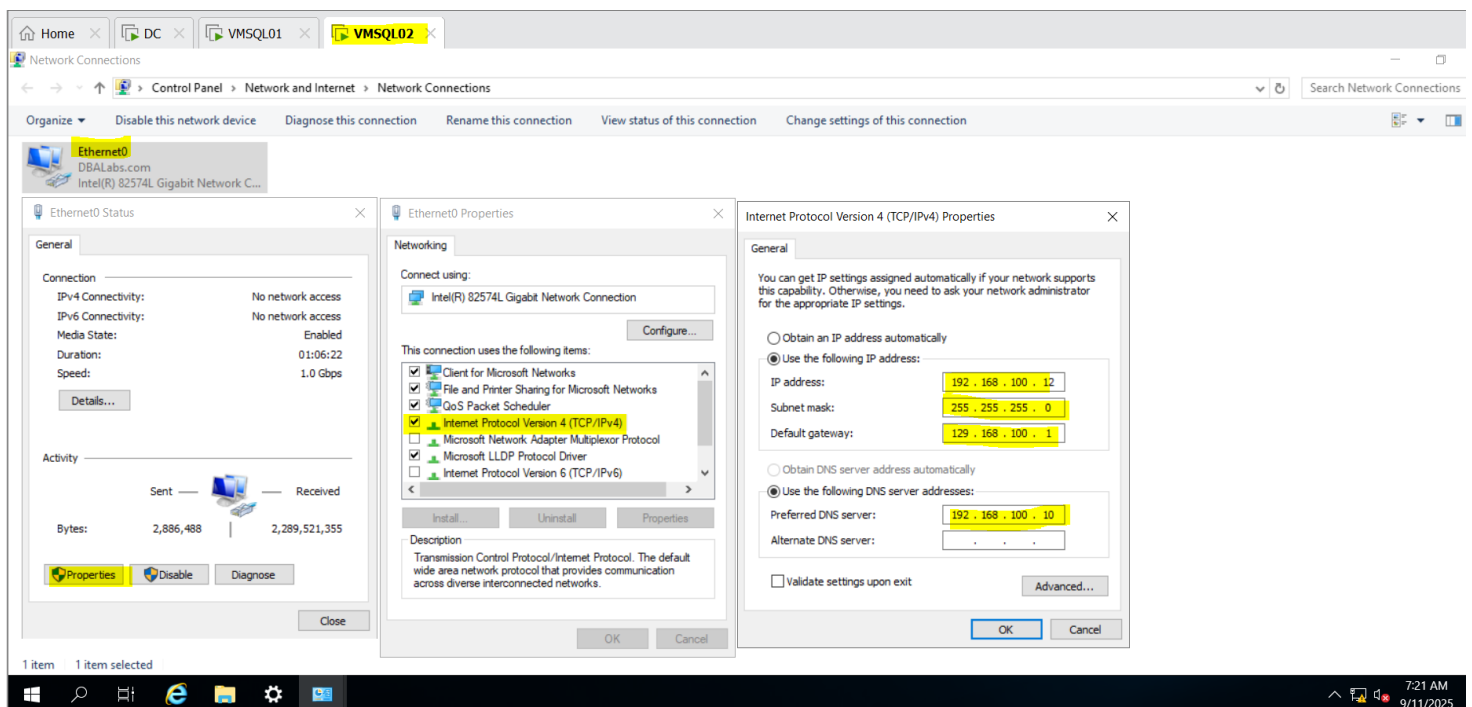
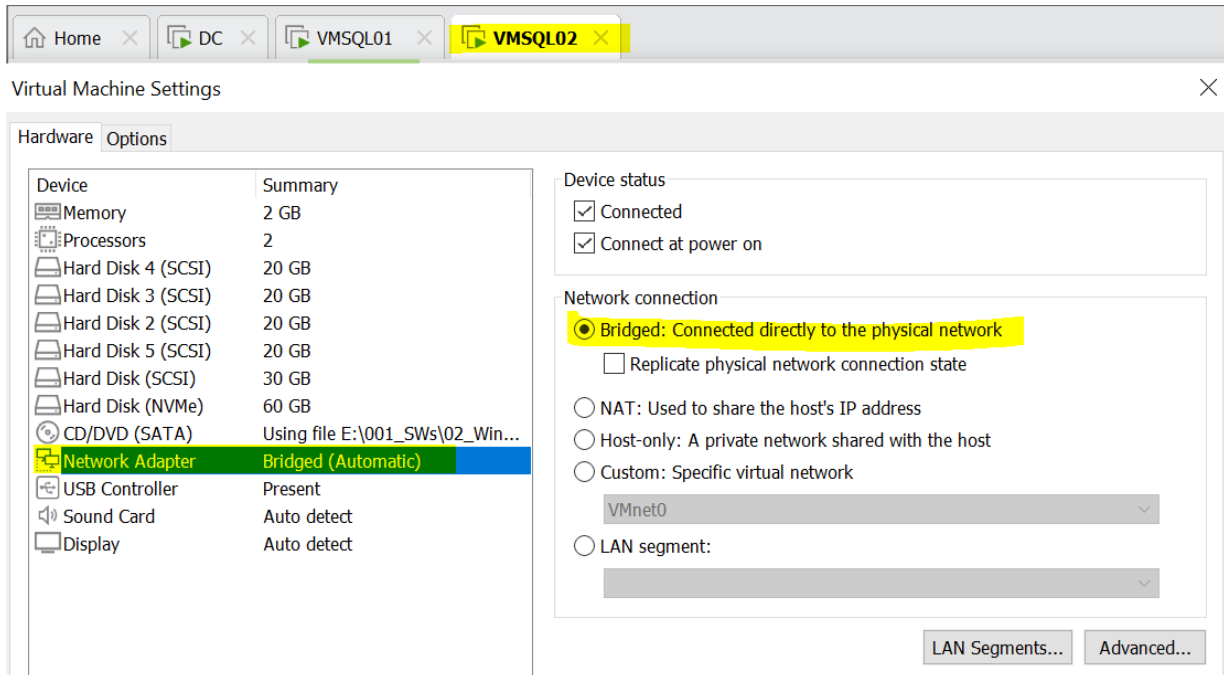
Remote Desktop

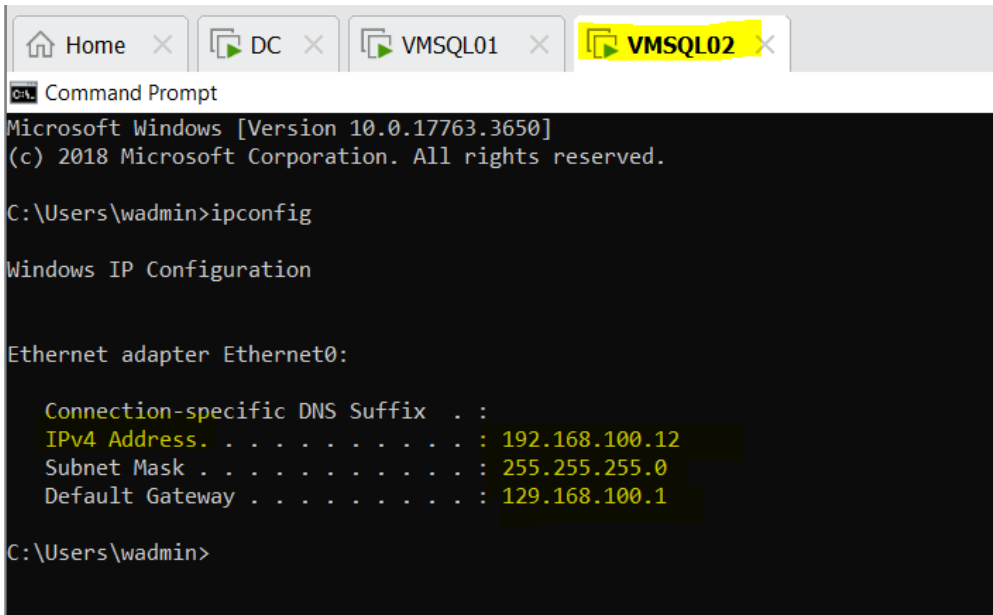
Choose an option, and then specify who can connect.

☐ Don't allow remote connections to this computer

☒ Allow remote connections to this computer

☐ Allow connections only from computers running Remote Desktop with Network Level Authentication (recommended)

At Node2 (VMSQL02):



The screenshot shows a Windows taskbar with four open applications: Home, DC, VMSQL01, and VMSQL02. The VMSQL02 application is highlighted in yellow. Below the taskbar, a Command Prompt window is open, displaying the output of the 'ipconfig' command. The output shows the IP configuration for the Ethernet adapter Ethernet0, including the IPv4 address, subnet mask, and default gateway.

```
Microsoft Windows [Version 10.0.17763.3650]
(c) 2018 Microsoft Corporation. All rights reserved.

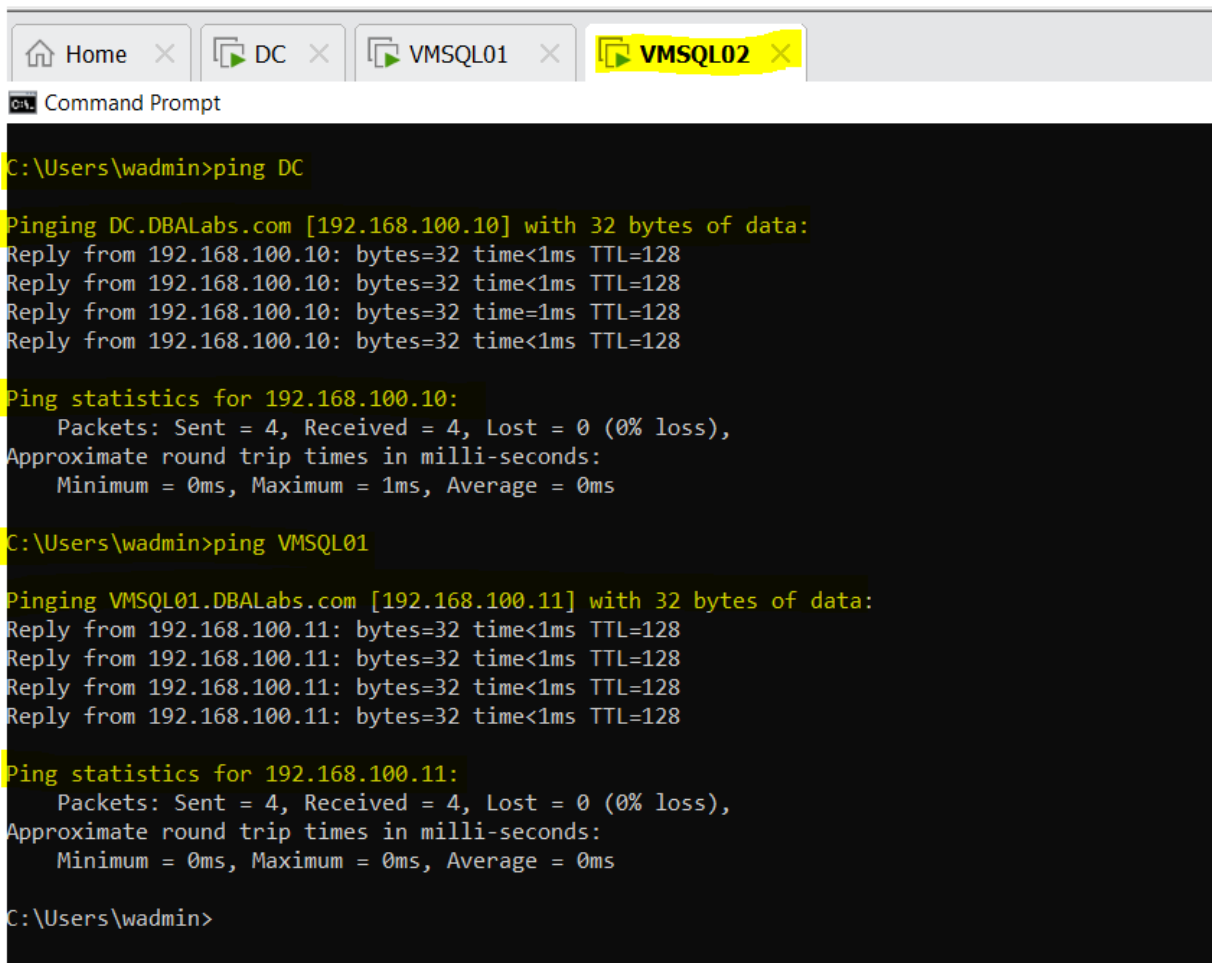
C:\Users\wadmin>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : 
    IPv4 Address. . . . . : 192.168.100.12
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 129.168.100.1

C:\Users\wadmin>
```



The screenshot shows a Windows taskbar with four open applications: Home, DC, VMSQL01, and VMSQL02. The VMSQL02 application is highlighted in yellow. Below the taskbar, a Command Prompt window is open, displaying the output of the 'ping DC' and 'ping VMSQL01' commands. The output shows the ping statistics for both targets, including the number of packets sent, received, and lost, and the approximate round trip times.

```
C:\Users\wadmin>ping DC

Pinging DC.DBALabs.com [192.168.100.10] with 32 bytes of data:
Reply from 192.168.100.10: bytes=32 time<1ms TTL=128
Reply from 192.168.100.10: bytes=32 time<1ms TTL=128
Reply from 192.168.100.10: bytes=32 time=1ms TTL=128
Reply from 192.168.100.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.100.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\wadmin>ping VMSQL01

Pinging VMSQL01.DBALabs.com [192.168.100.11] with 32 bytes of data:
Reply from 192.168.100.11: bytes=32 time<1ms TTL=128
Reply from 192.168.100.11: bytes=32 time<1ms TTL=128
Reply from 192.168.100.11: bytes=32 time<1ms TTL=128
Reply from 192.168.100.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.100.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\wadmin>
```

Home x DC x VMSQL01 x VMSQL02 x

Customize Settings

Control Panel > System and Security > Windows Defender Firewall > Customize Settings

Customize settings for each type of network

You can modify the firewall settings for each type of network that you use.

Domain network settings

☒ Turn on Windows Defender Firewall

- ☐ Block all incoming connections, including those in the list of allowed apps
- ☐ Notify me when Windows Defender Firewall blocks a new app

☒ Turn off Windows Defender Firewall (not recommended)

Private network settings

☒ Turn on Windows Defender Firewall

- ☐ Block all incoming connections, including those in the list of allowed apps
- ☐ Notify me when Windows Defender Firewall blocks a new app

☒ Turn off Windows Defender Firewall (not recommended)

Public network settings

☒ Turn on Windows Defender Firewall

- ☐ Block all incoming connections, including those in the list of allowed apps
- ☐ Notify me when Windows Defender Firewall blocks a new app

☒ Turn off Windows Defender Firewall (not recommended)

Home x DC x VMSQL01 x VMSQL02 x

System

Control Panel > System and Security > System

Control Panel Home

- Device Manager
- Remote settings
- Advanced system settings

View basic information about your computer

Windows edition

Windows Server 2019 Datacenter Evaluation
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System

Processor: 11th Gen Intel(R) Core(TM)
Installed memory (RAM): 2.00 GB
System type: 64-bit Operating System, x64
Pen and Touch: No Pen or Touch Input is available

Computer name, domain, and workgroup settings

Computer name: VMSQL02
Full computer name: VMSQL02.DBALabs.com
Computer description:
Domain: DBALabs.com

Windows activation

System Properties

Computer Name Hardware Advanced Remote

Remote Assistance

☐ Allow Remote Assistance connections to this computer

Advanced...

Remote Desktop

Choose an option, and then specify who can connect.

☐ Don't allow remote connections to this computer

☒ Allow remote connections to this computer

☐ Allow connections only from computers running Remote Desktop with Network Level Authentication (recommended)

Help me choose Select Users...